

Office VLAN Network – Technical Documentation

Table of contents

Contents	Page
Devices and Their Mapping	1
- Devices & Roles	
- Switch Port Mapping & VLAN Assignment	
VLAN & DHCP Summary	2
Router-on-a-Stick & DHCP Details	3
- ROAS Subinterfaces	
- DHCP Pools	
Wireless Access Points	4
Verification and Testing	5
- Verification Commands	

Devices & Their Mapping

Devices & Roles:

Device Name	Device Type	Role
R-CORE	Router 2811	Inter-VLAN Routing & DHCP
SW-CORE	Switch 2960	VLAN segmentation & trunking
AP-ADMIN-1	Access Point	VLAN 10 wireless
AP-FIN-1	Access Point	VLAN 20 wireless
AP-CS-1	Access Point	VLAN 30 wireless
PCs/Laptops/Mobile/Printers	End Devices	Connected to respective VLANs

Switch Port Mapping & VLAN Assignment:

Switch	Port	VLAN	Connected Device
SW-CORE	Fa0/1	Trunk	Link to R-CORE
SW-CORE	Fa0/2	10	Admin PC
SW-CORE	Fa0/3	10	Admin Printer
SW-CORE	Fa0/4	10	AP-ADMIN-1
SW-CORE	Fa0/5	20	Finance PC
SW-CORE	Fa0/6	20	Finance Printer
SW-CORE	Fa0/7	20	AP-FIN-1
SW-CORE	Fa0/8	30	Customer PC
SW-CORE	Fa0/9	30	Customer Printer
SW-CORE	Fa0/10	30	AP-CS-1

VLAN & DHCP Summary

VLAN ID	Department	Subnet	Gateway	DHCP Pool	Devices Connected
10	Admin	192.168.1.0/26	192.168.1.1	AdminPool	1 PC, 1 Laptop, 1 Printer, 1 AP
20	Finance	192.168.1.64/26	192.168.1.65	FinancePool	1 PC, 1 Laptop, 1 Printer, 1 AP
30	Customer Service	192.168.1.128/26	192.168.1.129	CustomerPool	1 PC, 1 Mobile, 1 Printer, 1 AP

Router-on-a-Stick & DHCP Details

ROAS Subinterfaces:

Subinterface	VLAN ID	IP Address	Subnet Mask
Fa0/0.10	10	192.168.1.1	255.255.255.192
Fa0/0.20	20	192.168.1.65	255.255.255.192
Fa0/0.30	30	192.168.1.129	255.255.255.192

DHCP Pools:

Pool Name	VLAN/Subnet	Default Gateway
AdminPool	192.168.1.0/26	192.168.1.1
FinancePool	192.168.1.64/26	192.168.1.65
CustomerPool	192.168.1.128/26	192.168.1.129

Wireless Access Points:

AP Name	VLAN	SSID	Security	Connected Devices
AP-ADMIN-1	10	ADMIN-WIRELESS	WPA2-PSK	Admin Laptop
AP-FIN-1	20	FIN-WIRELESS	WPA2-PSK	Finance Laptop
AP-CS-1	30	CS-WIRELESS	WPA2-PSK	Customer Mobile

Verification & Testing

Test Type	Result
Intra-VLAN ping	Successful
Inter-VLAN ping	Successful
DHCP assignment	Successful
Wireless connectivity	Successful

Verification Commands:

- ping <IP> on PCs / Laptops / Mobile
- ipconfig on end devices
- show ip interface brief on router